

Beyond Differential Gene Expression: The Case for Equivalence Testing RNA-Seq Data Analysis

Julia Olivieri, Assistant Professor of Computer Science at the University of the Pacific

Abstract

One of the fundamental questions of biology is how the overwhelming complexity of life on Earth can arise from our relatively simple genetic code. RNA sequencing (RNA seq) is one of the most common tools in a modern scientist’s toolkit to help answer this question. The most common analysis to perform on RNA-seq data is differential gene expression analysis: testing for statistically significant differences in gene expression between two or more groups, often incorporating an effect size cutoff.

For a given effect size (for example, 2x gene expression fold change), genes are divided into three groups: those with statistical evidence of a fold change greater than 2x, those with statistical evidence of a fold change $\leq 2x$, and those for which the statistical evidence is inconclusive. However, RNA-seq analysis does not distinguish between the latter two categories.

The field of equivalence testing is a field of hypothesis tests devoted to capturing statistical evidence that the difference in means between two groups is within a given interval, rather than greater than a value. Equivalence testing is commonly used in the fields of drug testing and consumer preference. Pairing tests of equivalence with tests of difference has the capacity to vastly expand the information content of our RNA-seq analyses.

Equivalence testing has the potential to advance many fields that rest on RNA-seq analysis. At a fundamental level, exploratory differential expression analyses attempt to build our understanding of gene functions by cataloguing in which circumstances the genes behave differently. However, understanding when the genes behave the same is just as important if we want a thorough understanding of the landscape of gene regulation. This would also allow us to focus additional resources on sequencing those genes in the “inconclusive” category, with the aim of developing a complete understanding of the mechanisms of gene regulation.

There are many situations in which reliably quantifying “equivalent genes” between two groups would provide meaningful information inaccessible by standard differential gene expression analysis. For example, understanding which genes are unaffected by specific pharmaceutical interventions would provide confidence that important pathways are not disrupted by a drug. For RNA-seq data from a biological sample with slightly mutated DNA vs a wild-type sample, the discovery of “equivalent” genes could let us draw conclusions about the limits of the effect of the mutation. Finding genes with “equivalent expression” between environmental conditions would help us understand which pathways were robust to specific environmental changes.

Equivalence testing between species has the potential to provide additional insights. Gene expression that is “equivalent” across species may be conserved due to functional importance. Also, equivalence testing could also help predict whether a discovery in a model organism will apply to humans. For example, if a drug successfully treated a mouse for a specific disease, and the genetic signatures of the disease in a mouse and human were equivalent, it could provide more confidence that the discovery would transfer to humans.

Equivalence testing could also help shed light on other questions in RNA biology, such as the prevalence of regulation of gene expression or splicing between different conditions [15], or the similarity of expression between separate batches sequences from the same population [19].

Thousands of RNA-seq experiments are made publicly available per year [7]. This large resource provides an opportunity to better understand functional regulation of gene expression and splicing, and provides a wealth of data to apply equivalence testing methods to. These methods are applicable to every gene expression analysis currently being performed, and would serve to significantly increase the field of gene regulation’s

base of knowledge. Understanding which genes are not differentially expressed is just as important as understanding which are differentially expressed when we are trying to categorize gene function. Additionally, this would let us identify which genes were “inconclusive,” and focus on collecting more data to shed light on these examples specifically (rather than collecting more data that we do not expect to tell us anything new).

In this study, we used the TOST equivalence testing approach to quantify difference and equivalence in a variety of two-sample RNA-seq studies. We identify new discoveries made possible by equivalence testing, demonstrate the applicability for additional analysis, provide simulations of TOST equivalence testing for RNA-sequencing data to understand how it is expected to work, and provide a framework for similar analysis in the future.

This approach divides all genes into those “with evidence of statistical difference between groups” and “without evidence of statistical difference between groups.” However, it does not distinguish between two groups in the later category: genes with statistical evidence of a difference smaller than the given effect size, and genes for which the evidence is inconclusive.

1 Introduction

1.1 The current paradigm of RNA-seq analysis

One of the fundamental questions of biology is how the overwhelming complexity of life on Earth can arise from our relatively simple genetic code. Organisms that share very similar DNA can look and behave in vastly different manners. What makes a brain cell different from a heart cell? Why can a small DNA mutation cause healthy tissue to become cancerous? How can the same plant (with a static genetic composition) adapt to weather conditions throughout the changing seasons? Fundamentally, the goal is to develop a complete understanding of RNA sequencing and its regulation...

RNA-sequencing (RNA-seq) is one of the most important tools in a modern scientist’s toolkit for answering these questions. Through whole-genome mRNA sequencing, individual transcripts from a biological sample are captured and measured in proportion to their abundance, allowing for accurate estimates of the expressions of all genes.

The most common analysis to perform on RNA-seq data is **differential gene expression analysis** [18]. This involves comparing gene expression values among two (or more) groups to determine which genes are differentially expressed between the groups. For example, if RNA-seq was performed on a cancerous and non-cancerous tissue sample, differential gene expression analysis could determine a set of genes with high statistical evidence for having different expression levels between the two groups.

This analysis approach partitions genes into two groups: those with evidence of differential gene expression, and those without evidence of differential gene expression. However, it provides no further insight into this second group of genes. Namely, is evidence of differential gene expression lacking because there is truly no difference in gene expression between the groups, or does the difference exist, but there was not enough data to detect it (Figure 2)? Equivalence testing lets us distinguish between these two groups.

1.2 What is equivalence testing?

Equivalence tests are a variety of hypothesis tests where the null hypothesis is that the difference in mean between two groups is larger than some pre-specified value, and the alternative hypothesis is that it is closer to zero. Incorporating equivalence tests in addition to difference tests will divide genes into three categories: those that have evidence of differential expression, those that have evidence of a lack of differential expression (beyond a threshold Δ), and those for which the data is inconclusive (Figure 2).

Despite similar approaches being used in the fields of generic drug testing and consumer preference

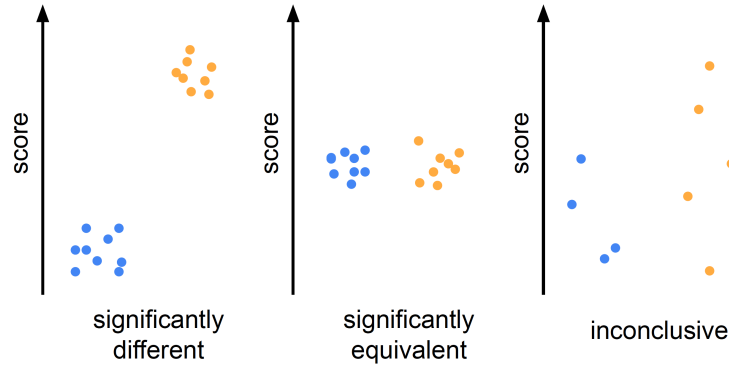


Figure 1: When testing whether the means are the same for a given population, these are the three possibilities. However, traditional significance testing does not distinguish between the second two cases.

[12], equivalence tests have very rarely been performed on gene expression data. When they have been, the goal has been to find reference genes suitable for normalization, rather than increasing our understanding of the biology itself [4, 3, 5]. Furthermore, I have yet to see an example of equivalence testing applied to RNA-sequencing data specifically, and have never seen equivalence testing applied to splicing data.

1.3 Why does equivalence testing matter?

2 Current tools and methods for equivalence testing

3 Equivalence testing examples

3.1 Case study: Equivalence testing in single-cell data

3.2 Case study: Equivalence testing for CRISPR screens

3.3 Case study: Equivalence testing between diseased and non-diseased samples

4 Unanswered questions

5 Conclusion

Equivalence Testing for RNA Sequencing Data

Julia Olivieri, Assistant Professor of Computer Science at the University of the Pacific

6 Background

One of the fundamental questions of biology is how the overwhelming complexity of life on Earth can arise from our relatively simple genetic code. Organisms that share very similar DNA can look and behave in vastly different manners. What makes a brain cell different from a heart cell? Why can a small DNA mutation cause healthy tissue to become cancerous? How can the same plant (with a static genetic composition) adapt to weather conditions throughout the changing seasons?

To understand how these differences occur, we need to know what information DNA encodes. The DNA sequence for an organism is divided into many subsections, called **genes**. Genes contain the instructions for creating **proteins**, which are like the molecular machines of the cell. Proteins perform all of the jobs necessary for a cell to function properly. Depending on which proteins are expressed in different cells, and the number of copies of each protein present in those cells, the cells can behave very differently.

Protein levels are difficult to measure, but a technique called **RNA sequencing (RNA-seq)** quantifies how many copies of each “set of instructions” (RNA molecules) are present in a given biological sample. This sample could be almost any biological specimen: leaf, a cancerous tumor, or even a single cell. The amount of each RNA molecule present in a given biological sample serves as an approximation of the amount of each individual protein present in that sample. We summarize this information as “**gene expression**” data: the number of copies of each gene that have been sequenced from the biological sample. RNA-seq data can also provide information about RNA splicing [14] or polyadenylation [11], which determine which exact protein version is created from the given gene.

The most common analysis to perform on RNA-seq data is **differential gene expression analysis** [18]. This involves comparing gene expression values among two (or more) groups to determine which genes are differentially expressed between the groups. For example, if RNA-seq was performed on a cancerous and non-cancerous tissue sample, differential gene expression analysis could determine a set of genes with high statistical evidence for having different expression levels between the two groups.

Differential expression analysis is usually accomplished by performing a separate significance test for each gene (followed by multiple hypothesis correction of the p values). Genes with the lowest p values are categorized as “significantly different.” However, no conclusion can be drawn about the remaining genes. There are two possibilities: Either there is no difference in gene expression between the groups, or there is a difference between two of the groups but there was not enough data to detect it.

The goal of this project is to apply methods from the field of **equivalence testing** develop an easy-to-use procedure to help distinguish between these latter two cases [12]. This will divide genes into three categories: those that have evidence of differential expression, those that have evidence of a lack of differential expression (beyond a threshold Δ), and those for which the data is inconclusive (Figure 2).

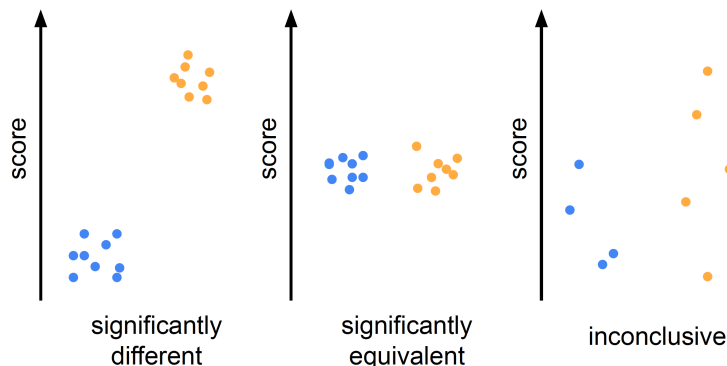


Figure 2: When testing whether the means are the same for a given population, these are the three possibilities. However, traditional significance testing does not distinguish between the second two cases.

Despite similar approaches being used in the fields of generic drug testing and consumer preference [12], equivalence tests have very rarely been performed on gene expression data. When they have been, the goal has been to find reference genes suitable for normalization, rather than increasing our understanding of the biology itself [4, 3, 5]. Furthermore, I have yet to see an example of equivalence testing applied to RNA-sequencing data specifically, and have never seen equivalence testing applied to splicing data.

Equivalence testing has the potential to advance many fields that rest on RNA-seq analysis. At a fundamental level, exploratory differential expression analyses attempt to build our understanding of gene functions by cataloguing in which circumstances the genes behave differently. However, understanding when the genes behave the same is just as important if we want a thorough understanding of the landscape of gene regulation. This would also allow us to focus additional resources on sequencing those genes in the “inconclusive” category, with the aim of developing a complete understanding of the mechanisms of gene regulation.

There are many situations in which reliably quantifying “equivalent genes” between two groups would provide meaningful information inaccessible by standard differential gene expression analysis. For example, understanding which genes are unaffected by specific pharmaceutical interventions would provide confidence that important pathways are not disrupted by a drug. For RNA-seq data from a biological sample with slightly mutated DNA vs a wild-type sample, the discovery of “equivalent” genes could let us draw conclusions about the limits of the effect of the mutation. Finding genes with “equivalent expression” between environmental conditions would help us understand which pathways were robust to specific environmental changes.

Equivalence testing between species has the potential to provide additional insights. Gene expression that is “equivalent” across species may be conserved due to functional importance. Also, equivalence testing could also help predict whether a discovery in a model organism will apply to humans. For example, if a drug successfully treated a mouse for a specific disease, and the genetic signatures of the disease in a mouse and human were equivalent, it could provide more confidence that the discovery would transfer to humans.

Equivalence testing could also help shed light on other questions in RNA biology, such as the prevalence of regulation of gene expression or splicing between different conditions [15], or the similarity of expression between separate batches sequences from the same population [19].

Thousands of RNA-seq experiments are made publicly available per year [7]. This large resource provides an opportunity to better understand functional regulation of gene expression and splicing, and provides a wealth of data to apply equivalence testing methods to. These methods are applicable to every gene expression analysis currently being performed, and would serve to significantly increase the field of gene regulation’s base of knowledge. Understanding which genes are not differentially expressed is just as important as understanding which are differentially expressed when we are trying to categorize gene function. Additionally, this would let us identify which genes were “inconclusive,” and focus on collecting more data to shed light on these examples specifically (rather than collecting more data that we do not expect to tell us anything new).

Additionally, this work would support interdisciplinary initiatives within the University of the Pacific. It combines aspects of computer science, biology, and statistics. There are opportunities for collaboration with undergraduate or graduate students interested in any of these fields. On top of that, this project involves significant data analysis, which synergizes well with the new data science major that was just approved. Overall, this project has the potential to contribute to the research goals of the University of the Pacific as a whole.

7 Rationale, Approach, Objectives

Project objectives:

1. **Equivalence testing implementation:** Implement a method that can robustly test for “equivalence”

between two (or more) conditions of RNA-seq data.

2. **Data mining of previously-published data for equivalent gene discovery:** Apply this method to a wide variety of existing data to create an atlas of equivalent gene behavior.

Equivalence testing implementation

Technical problem description

Given two populations f and g , it is standard for a significance test to be performed to test whether the two population means, μ_f and μ_g , are significantly different. In our case, the two populations would be two different RNA-seq samples, and for a specific gene μ_f is the mean gene expression value (or splicing value) for the gene in population f , and μ_g is the mean gene expression value (or splicing value) for the gene in population g .

In this case, the null hypothesis would be that the population means are the same ($H_0 : \mu_f = \mu_g$), and the alternative hypothesis would be that the population means are different ($H_1 : \mu_f \neq \mu_g$).

Rejecting the null hypothesis means that there is significant evidence that the two population means are different (specifically, the probability of observing samples with a difference at least as large as what was observed is low under the null hypothesis). If we fail to reject the null hypothesis, there are two possibilities: either the two population means are the same ($\mu_f = \mu_g$), or the two population means are different ($\mu_f \neq \mu_g$), but the samples do not provide strong enough evidence to draw this conclusion.

If we are interested in testing for equivalence rather than testing for difference, we would ideally want to test the opposite: Let the null hypothesis be that the population means are different ($H_0 : \mu_f \neq \mu_g$), and the alternative hypothesis be that the two population means are the same ($H_1 : \mu_f = \mu_g$). But it is impossible to reject this null hypothesis: the two population means could always be very slightly different, beyond the limit of detection by the current samples. Moreover, we are usually not interested in differences between population means that are miniscule.

Instead, in the field of equivalence testing, the null hypothesis under consideration is that the difference between the two population means is greater than some threshold Δ , ($H_0 : |\mu_f - \mu_g| > \Delta$), and the alternative hypothesis is that the difference between population means is less than or equal to Δ , ($H_1 : |\mu_f - \mu_g| \leq \Delta$). These are the hypotheses that we will test. Specifically, considering one gene and two biological samples f and g , we are interested in testing whether the mean gene expression values (or splicing values) of the two populations are within Δ of each other.

7.0.1 Choosing Δ

The choice of Δ is critical for performing a valid equivalence test. If Δ is too large, then a difference $\leq \Delta$ in population means does not imply that the populations are functionally equivalent. If Δ is too small, we lose power to detect functionally equivalent populations.

Intuitively, if the difference in means between populations is within the range of differences found within one of the populations, the populations could be thought of as functionally equivalent. Within each population, we expect there to be some variation in score. It is intuitive to understand that differences commonly found between samples in a single population should be within the allowable range between populations while still considering them to be functionally equivalent. Therefore, we will find the interquartile range of each sample, and let Δ equal the smaller of these two interquartile ranges. This is a conservative approach, because it results in a smaller threshold than if we took the average or maximum of these ranges.

I will test and modify this definition of Δ based on simulations. Specifically, I will simulate samples between two populations using negative binomial distributions, the preferred distribution for modeling RNA-seq data [16]. I will quantify how this equivalence test performs as the variance and difference between

populations changes. I will also tune Δ by testing two experimental samples drawn from the same population against each other, with the understanding that the genes in those two samples should be functionally equivalent. Δ will be modified such that its results are in concordance with biological studies.

7.0.2 Equivalence testing procedure

Once Δ has been determined, I will use the two one-sided tests procedure (TOST) to determine whether the null hypothesis ($H_0 : |\mu_f - \mu_g| > \Delta$) can be rejected in favor of the alternative hypothesis ($H_1 : |\mu_f - \mu_g| \leq \Delta$) (Figure 3). To perform TOST, two one-sided, one-sample hypothesis test are performed. The first uses a one-sample t-test to test the following two hypotheses:

$$H_0^{(1)} : \mu_f - \mu_g < -\Delta$$

$$H_1^{(1)} : \mu_f - \mu_g \geq -\Delta$$

The second uses a one-sample t-test to test the complementary hypotheses:

$$H_0^{(2)} : \mu_f - \mu_g > \Delta$$

$$H_1^{(2)} : \mu_f - \mu_g \leq \Delta$$

The resulting p value is the maximum of the two p values resulting from these two tests. Even though we are performing two hypothesis tests here, we do not need to perform multiple hypothesis testing correction because we only reject the null hypothesis if both of the tests are below our chosen significance level α .

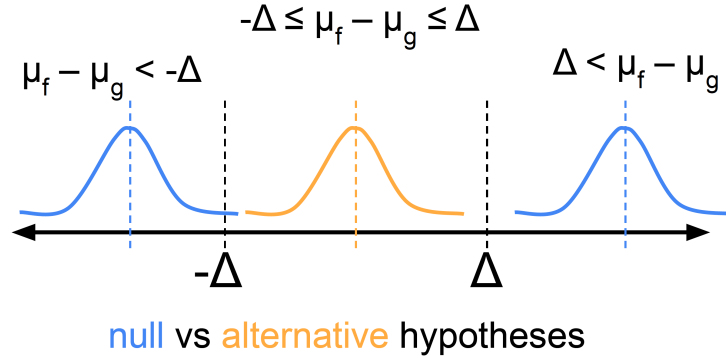


Figure 3: Null hypotheses (in blue) vs alternative hypotheses (in orange) for TOST procedure

This approach will be used because it is intuitive, it is a valid α -level test, and it is unbiased for reasonable sample sizes [12]. The assumptions required for this test to be valid are that the samples are continuous, approximately normally distributed, and there should be no significant outliers. The scores we will be examining are continuous, and are expected to be approximately normally distributed for reasonable sample sizes. Data will be pre-process to remove significant outliers.

7.0.3 Details of the TOST implementation

The two t statistics tested in the TOST procedure are the following:

$$t_1 = \frac{(\mu_f - \mu_g) - \Delta}{\sqrt{\frac{s_f^2}{n_f} + \frac{s_g^2}{n_g}}}$$

$$t_2 = \frac{(\mu_f - \mu_g) + \Delta}{\sqrt{\frac{s_f^2}{n_f} + \frac{s_g^2}{n_g}}}$$

Where s_f is the sample standard deviation of population f , s_g is the sample standard deviation of population g , n_f is the number of measurements from sample f , and n_g is the number of measurements from population g .

Then the resulting p value is $\max(\text{cdf}(t_1, \nu), 1 - \text{cdf}(t_2, \nu))$.

Here “cdf” is the cdf of Student’s t distribution. The degrees of freedom are $n_f + n_g - 1$.

NOTE: I think this is wrong - the degrees of freedom should be approximated from the Welch–Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_f^2}{n_f} + \frac{s_g^2}{n_g}\right)^2}{\frac{s_f^4}{n_f^2(n_f-1)} + \frac{s_g^4}{n_g^2(n_g-1)}}$$

7.0.4 Vectorization

The t statistic for Welch’s two-sided two-sample t test for difference of means is:

Reference how it’s commonly advised to use Welch in the vast majority of cases.

$$t = \frac{\mu_f - \mu_g}{\sqrt{\frac{s_f^2}{n_f} + \frac{s_g^2}{n_g}}}$$

The resulting p value is $2\text{cdf}(t, \nu)$, where $\mu_f \leq \mu_g$.

In this case, the degrees of freedom ν can’t only be calculated as $n_1 + n_2 - 2$, because we’re not assuming the variances of the two populations are the same. Instead, the degrees of freedom are approximated by:

$$\nu \approx \frac{\left(\frac{s_f^2}{n_f} + \frac{s_g^2}{n_g}\right)^2}{\frac{s_f^4}{n_f^2(n_f-1)} + \frac{s_g^4}{n_g^2(n_g-1)}}$$

7.0.5 Effect size cutoff

We are interested in comparing differential expression to equivalent expression. For testing equivalent expression, we are required to choose a Δ value to bound what we mean by “equivalent.” This is not required for differential expression. However, many differential expression analyses include an effect size cutoff for significant results as well, because differences are usually only interesting when their magnitude is large enough (and p values can be arbitrarily small even for tiny differences given enough data).

Therefore, for each gene we will only test for equivalence if the difference in means is $> \Delta$, and test for equivalence if the difference in means is $\leq \Delta$. Note that filtering based on effect size can bias multiple hypothesis testing results [21], so we will compute p values for all and then correct them before filtering on effect size.

7.0.6 Equivalence testing for fold change

Often, cutoffs for differential RNA sequencing are determined by the fold change difference, with a fold change difference of 2 being a common cutoff (e.g., in order for two genes to be determined to be differentially expressed, one must be expressed at least twice as highly as the other) [10]. In our case, this translates to:

$$\frac{1}{2} \leq \frac{\mu_f}{\mu_g} \leq 2$$

If we take the log of each expression, this gives us:

$$-1 \leq \log_2(\mu_f) - \log_2(\mu_g) \leq 1$$

Therefore, we can do our same equivalence testing procedure with $\Delta = 1$ on $\log_2$ -transformed data to calculate fold-change equivalence. This is interpretable based on the common “volcano plot,” which has $\log_2$ fold change on the x axis and $-\log_{10}$ p value on the y axis.

7.0.7 Equivalence testing with weighted means

When applying TOST to RNA-seq data, there is one important consideration to keep in mind: the confidence we have in each observation is not consistent. Each one of our observations represents a gene measurement from an RNA-seq experiment. But if one of these experiments is deeply sequenced with 5,000,000 reads, and another was shallowly sequenced with 10,000 reads, we would have more confidence in the score for the first experiment.

We can modify the TOST procedure to account for this by weighting our sample means by the number of reads that contributed to each score (the number of total reads in an experiment for a gene expression analysis, and the number of reads from the gene for a splicing analysis). Then the equivalence testing procedure described above can be performed using the different of weighted means, with the corresponding weighted standard deviations.

Note that if the sampling depths across experiments are relatively consistent, this would not be expected to have a large effect on the results. However, it could help mitigate outliers due to insufficient sequencing if the depths are different, as is often the case in single-cell RNA-seq studies.

7.0.8 Implementing equivalence testing software

The equivalence testing procedure described above will be implemented in a python software package, with an emphasis on making it easy-to-use for other scientists. This pipeline will take as input the output of common differential-expression analysis software like DEseq-2 [9] and splicing analysis software like the SpliZ [14], and output the results of equivalence testing on this data according to the procedure outlined above.

7.1 Data mining of previously-published data for equivalent gene discovery

Once I have implemented equivalence testing for RNA-seq experiments, I will start applying it to existing datasets to address fundamental open questions in the field of gene regulation. Although there are many potential applications, I will first focus on four applications that showcase the range of questions that can be addressed using this method.

For each of these applications, I will download the previously-published RNA-sequencing data on to Pacific’s computational cluster, quantify gene expression or splicing levels as appropriate, and apply my

analysis pipeline to identify genes with “equivalent” gene expression or splicing, as well as a range of genes with differential expression or differential splicing regulation.

7.1.1 Identifying genes unaffected by drought conditions in grapes

As climate change continues to reshape our world, understanding the way changing environmental conditions affect crops is essential for ensuring that global agriculture is prepared for the changes ahead. Specifically, droughts are expected to be increasingly common in the coming years [13]. There has been significant investigation of differential gene expression changes in crops due to drought conditions [17]. However, there has not been an investigation of genes with expression unaffected by drought conditions. I will analyze equivalent gene expression in wine-producing grapes using data from the study [2]. I will compare the gene expression of the “unstressed” roots to the “stressed” roots to identify genes that are robust to water-stressed conditions, providing evidence for pathways that can be expected to be unaffected by future drought conditions.

7.1.2 Identifying genes unaffected by FOXC2 disruption in B16-F1 melanoma

CRISPR technology has enabled the study of specific mutations, allowing for a more precise understanding of gene regulation than ever before [1]. Differential gene expression analysis of CRISPR data has helped elucidate the gene regulatory networks. However, these analyses do not quantify the genes that are unaffected by these mutations. I will use data from a CRISPR study involving analysis of FOXC2-deficient melanoma cells compared to wild-type melanoma cells using data from [6]. FOXC2 is known to regulate tumor progression in many cell types, so understanding genes with unaffected expression levels by this mutation can provide a bound on the secondary effects of these mutations, and help localize the genes that are affected.

7.1.3 Quantifying batch effects between Smart-seq2 and 10X Chromium data in the Human Lung Cell Atlas

As single-cell RNA-sequencing studies continue to grow in number and size, researchers are increasingly facing the challenge of how to “correct” batch effects between data collected by different technologies. More specifically, many single-cell sequencing studies collect both Smart-seq2 data, which allows for deep, full-length sequencing of individual cells but is difficult to scale, and 10X Chromium data, which is easy to collect for many thousands of cells but is sparse and 3'-biased. Many methods exist to attempt to correct batch effects between these data types. If the two data types provided equivalent data, we would expect high gene expression equivalence between them. I plan on using this equivalence testing framework to quantify batch differences between Smart-seq2 and 10X Chromium data from the Human Lung Cell Atlas, [20]. This will allow for the comparison of different batch effect methods, and analysis of how extremely they are needed.

7.1.4 Estimating the extent of splicing regulation in pollen genesis

The extent of splicing regulation, and specifically the amount of splicing that is functional versus biological noise, is hotly debated in the splicing field [22]. To provide an estimate of the fraction of splicing that is regulated in one specific condition, I will quantify arabidopsis splicing in two stages of pollen differentiation from [8]. This will provide an upper and lower bound on splicing regulation captured in this specific differentiation stage, and will provide a blueprint of how this question can be addressed for other areas of interest. I have completed some preliminary analysis on this dataset, which indicates that equivalent genes are found between stages of spermatogenesis (Figure 4).

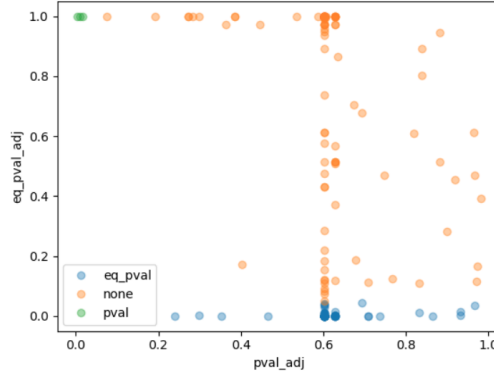


Figure 4: Preliminary data shows the relationship between the p value from differential expression analysis (x axis) and equivalence analysis (y axis). A substantial number of genes (in blue) were found to have equivalent expression between two stages of pollenogenesis.

7.2 Extensions of this work

A robust, easy-to-use software package to implement equivalence testing for commonly-used RNA-seq scores has the potential to become widely used throughout the transcriptomics field. Any differential analysis of scores between subgroups from RNA-seq data could be extended to also provide those scores with evidence of score equivalence. In understanding gene regulation, we have so far focused too much on difference at the expense of similarity.

Additionally, as new data types arise and large, sparse data continues to be common, this method will be able to answer currently-open questions surrounding the prevalence of regulation of gene expression and splicing, and whether higher sequencing depths would be expected to yield new insights.

There are many directions in which this work could be extended. Rather than focusing on two-sample significance testing, the method could be expanded to assess an arbitrary number of groups. Also, The method could be modified to quantify equivalence within a difference of a fold change, rather than equivalence within an additive difference. This would be valuable because gene expression changes are often quantified in terms of fold changes [9].

Overall, this project would be a jumping-off point for the central question of my lab: How can we best leverage RNA-seq data to understand gene regulation? More than that, it has the potential to be transformative for how standard analyses are performed in the field, adding another tool to the computational biologist’s toolkit.

8 Timeline

This project will be completed in two distinct phases, with goals for papers and grants corresponding to each phase, as well as planned collaboration with students for each phase.

Equivalence testing implementation

1. *Nov 2023 - May 2024*: Preliminary implementation and verification of equivalence testing for RNA-seq.
2. *Jun 2024 - Aug 2024*: Work with a summer student to refine this implementation and verification, and prepare for publication.
3. *Aug 2024*: Submit paper on the equivalence testing for RNA-seq method to *Genome Biology*.

4. *Sep 2024*: Apply for NSF CRII grant to support this project.

Data mining of previously-published data for equivalent gene discovery

5. *Oct 2024 - Apr 2025*: Begin running the completed algorithm on available RNA-seq datasets and analyzing the results.
5. *May 2025 - July 2025*: Work with a summer student to complete data mining and analysis, and summarize these biological findings for publication.
5. *Jul 2025*: Submit paper on data analysis results to *eLife*.
5. *Jul 2025*: Apply for NSF CAREER grant to support this project.

References

- [1] Rodolphe Barrangou and Jennifer A Doudna. Applications of crispr technologies in research and beyond. *Nature biotechnology*, 34(9):933–941, 2016.
- [2] Massimiliano Corso, Alessandro Vannozzi, Elie Maza, Nicola Vitulo, Franco Meggio, Andrea Pitacco, Andrea Telatin, Michela D’angelo, Erika Feltrin, Alfredo Simone Negri, et al. Comprehensive transcript profiling of two grapevine rootstock genotypes contrasting in drought susceptibility links the phenylpropanoid pathway to enhanced tolerance. *Journal of experimental botany*, 66(19):5739–5752, 2015.
- [3] Emmanuel Curis, Calypso Nepost, Diane Grillault Laroche, Cindie Courtin, Jean-Louis Laplanche, Bruno Etain, and Cynthia Marie-Claire. Selecting reference genes in rt-qpcr based on equivalence tests: A network based approach. *Scientific Reports*, 9(1):16231, 2019.
- [4] Wouter J Eijgelaar, Anton JG Horrevoets, Ann-Pascale JJ Bijnens, Mat JAP Daemen, and Wim FJ Verhaegh. Equivalence testing in microarray analysis: similarities in the transcriptome of human atherosclerotic and nonatherosclerotic macrophages. *Physiological genomics*, 41(3):212–223, 2010.
- [5] Florian Haller, Bettina Kulle, Stefanie Schwager, Bastian Gunawan, Anja von Heydebreck, Holger Sülthmann, and László Füzesi. Equivalence test in quantitative reverse transcription polymerase chain reaction: confirmation of reference genes suitable for normalization. *Analytical biochemistry*, 335(1):1–9, 2004.
- [6] Kristian M Hargadon and Corey J Williams. Rna-seq analysis of wild-type vs. foxc2-deficient melanoma cells reveals a role for the foxc2 transcription factor in the regulation of multiple oncogenic pathways. *Frontiers in Oncology*, 10:267, 2020.
- [7] Alexander Lachmann, Denis Torre, Alexandra B Keenan, Kathleen M Jagodnik, Hoyjin J Lee, Lily Wang, Moshe C Silverstein, and Avi Ma’ayan. Massive mining of publicly available rna-seq data from human and mouse. *Nature communications*, 9(1):1366, 2018.
- [8] Liam Le Lievre, Sreejith P Chakkatu, Shiny Varghese, Robert C Day, Sarah M Pilkington, and Lynette Brownfield. Rna-seq analysis of synchronized developing pollen isolated from a single anther. *Frontiers in Plant Science*, 14:1121570, 2023.
- [9] Michael I Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome biology*, 15(12):1–21, 2014.
- [10] Davis J McCarthy and Gordon K Smyth. Testing significance relative to a fold-change threshold is a treat. *Bioinformatics*, 25(6):765–771, 2009.

- [11] Elisabeth Meyer, Kaitlin Chaung, Roozbeh Dehghannasiri, and Julia Salzman. Readzs detects cell type-specific and developmentally regulated rna processing programs in single-cell rna-seq. *Genome Biology*, 23(1):1–28, 2022.
- [12] Michael Meyners. Equivalence tests—a review. *Food quality and preference*, 26(2):231–245, 2012.
- [13] Sourav Mukherjee, Ashok Mishra, and Kevin E Trenberth. Climate change and drought: a perspective on drought indices. *Current climate change reports*, 4:145–163, 2018.
- [14] Julia Eve Olivieri, Roozbeh Dehghannasiri, and Julia Salzman. The spliz generalizes ‘percent spliced in’ to reveal regulated splicing at single-cell resolution. *Nature methods*, 19(3):307–310, 2022.
- [15] Julia Eve Olivieri, Roozbeh Dehghannasiri, Peter L Wang, SoRi Jang, Antoine De Morree, Serena Y Tan, Jingsi Ming, Angela Ruohao Wu, Stephen R Quake, Mark A Krasnow, et al. Rna splicing programs define tissue compartments and cell types at single-cell resolution. *Elife*, 10:e70692, 2021.
- [16] Xu Ren and Pei-Fen Kuan. Negative binomial additive model for rna-seq data analysis. *BMC bioinformatics*, 21:1–15, 2020.
- [17] Arun K Shanker, M Maheswari, SK Yadav, S Desai, Divya Bhanu, Neha Bajaj Attal, and B Venkateswarlu. Drought stress responses in crops. *Functional & integrative genomics*, 14:11–22, 2014.
- [18] Cynthia M Sharma and Jörg Vogel. Differential rna-seq: the approach behind and the biological insight gained. *Current opinion in microbiology*, 19:97–105, 2014.
- [19] Hoa Thi Nhu Tran, Kok Siong Ang, Marion Chevrier, Xiaomeng Zhang, Nicole Yee Shin Lee, Michelle Goh, and Jinmiao Chen. A benchmark of batch-effect correction methods for single-cell rna sequencing data. *Genome biology*, 21:1–32, 2020.
- [20] Kyle J Travaglini, Ahmad N Nabhan, Lolita Penland, Rahul Sinha, Astrid Gillich, Rene V Sit, Stephen Chang, Stephanie D Conley, Yasuo Mori, Jun Seita, et al. A molecular cell atlas of the human lung from single-cell rna sequencing. *Nature*, 587(7835):619–625, 2020.
- [21] Maarten van Iterson, Judith M Boer, and Renée X Menezes. Filtering, fdr and power. *BMC bioinformatics*, 11:1–11, 2010.
- [22] Yihan Wan and Daniel R Larson. Splicing heterogeneity: separating signal from noise. *Genome biology*, 19(1):1–10, 2018.